

# Flyber Data Strategy MVP

## Introduction

Flyber has been massively successful. Results have beaten expectations and projections! This is good news for Flyber, but now it's time to plan for what's next. With success came some challenges. While we were able to grow, the original data pipelines to receive and process data are unable to keep up with the current and future growth.

As a Data Product Manager, working with multiple teams and stakeholders is imperative to success. To understand what our needs are, what scale we are growing at, and how we can build for the future, we need to consider all relevant stakeholders. In this proposal, present your findings along with the analysis and reasoning behind the choices made in order to help Flyber continue its success.

## Section 1: Data Customers & Needs

Flyber is a two-sided platform. You have customers who are riders, and you have partners who are drivers/pilots (think Uber: riders and drivers). For the Minimum Viable Product, you will be focusing on the Riders side of the business. To build an end to end data pipeline the very first step is to understand who needs data and why they need that data. Within Flyber, identify who your primary data customers/stakeholders are, why they are your primary data stakeholders and how they want to use the data (primary use-cases).

### Identify your primary internal stakeholders and their use-cases:

*(You may add more rows if necessary.)*

| Stakeholder            | Why are they primary stakeholders?                                                                                                                                    | Use-Case                                                                                                                                            |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Engineering / Platform | Owns the live application, instrumentation, pipeline reliability, and capacity. It is primary because every other team depends on complete, timely, trustworthy data. | 1) Monitor event volume and ingestion failures to scale infrastructure. 2) Diagnose app and pipeline issues by device, page, location, and release. |
| Product Management     | Owns rider experience and prioritizes the MVP roadmap. It is primary because behavioral evidence is needed to identify friction and validate improvements.            | 1) Measure the rider funnel from open/search to request/begin ride. 2) Compare adoption and conversion by device, page, and neighborhood.           |

|                      |                                                                                                                                                        |                                                                                                                                       |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Marketing / Growth   | Owns customer acquisition and campaign investment. It is primary because campaigns directly change demand, event volume, and infrastructure load.      | 1) Attribute traffic and ride-intent changes to campaigns. 2) Compare campaign response by acquisition channel, location, and cohort. |
| Finance / Operations | Owns unit economics, forecasts, and operating capacity. It is primary because transactional growth drives revenue, service demand, and warehouse cost. | 1) Forecast ride demand and data-processing cost. 2) Reconcile request/begin-ride activity with operational and financial reporting.  |

## Section 2: Data Collection and Data Modelling

**To support our primary stakeholders's use-cases we need following data:**

*(You may add more rows if necessary.)*

| Stakeholder            | Use-Case                                              | Data                                                                                                               | Why is this the primary use-case?                                                                                                                        |
|------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Engineering / Platform | Capacity monitoring; incident diagnosis               | event_id, event_time, event_type, ingestion_time, pipeline_status; device_type, app_version, page_type, error_code | Timestamps and IDs quantify throughput and latency; status/error fields locate failures; device, release, and page dimensions isolate affected segments. |
| Product Management     | Funnel measurement; segment comparison                | rider_id, session_id, event_type, event_time; page_type, device_type, neighborhood, app_version                    | Rider/session keys preserve journeys; ordered event types measure conversion; product dimensions reveal where behavior differs.                          |
| Marketing / Growth     | Campaign attribution; cohort response                 | rider_id, campaign_id, acquisition_channel, event_time; event_type, neighborhood, device_type, experiment_variant  | Campaign/channel and time connect exposure to behavior; event and segment fields measure incremental response and support cohort comparisons.            |
| Finance / Operations   | Demand and cost forecasts; transaction reconciliation | ride_id, rider_id, request_time, begin_ride_time,                                                                  | Ride keys and states support auditable transaction counts; volume, processing,                                                                           |

|  |  |                                                                                     |                                                                   |
|--|--|-------------------------------------------------------------------------------------|-------------------------------------------------------------------|
|  |  | ride_status;<br>event_count,<br>bytes_processed,<br>neighborhood,<br>warehouse_cost | location, and cost fields support capacity and expense forecasts. |
|--|--|-------------------------------------------------------------------------------------|-------------------------------------------------------------------|

### The tables we need are:

*Note: As a best practice, we should establish these relationships between tables from the very beginning. To complete this exercise we will focus on fundamental concepts of relational databases - tables, normalization and unique keys. Please provide the table header row for each table, tables might be different lengths. Make sure you include the following for each table. You can create as many tables as you feel are necessary (copy and paste from one of the table sections):*

#### **Table 1:**

Rider

*(You may add more columns if necessary.)*

|               |            |                   |                     |                |             |
|---------------|------------|-------------------|---------------------|----------------|-------------|
| rider_id (PK) | created_at | home_neighborhood | acquisition_channel | consent_status | campaign_id |
|---------------|------------|-------------------|---------------------|----------------|-------------|

Rationale for Choosing Primary and Foreign Keys for the Table 1:

rider\_id is a stable surrogate identifier that avoids mutable or sensitive natural keys such as email or phone. campaign\_id is nullable because organic riders have no campaign. This table stores one row per rider, preventing repeated rider attributes in every event.

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#### **Table 2:**

Session

*(You may add more columns if necessary.)*

|                 |               |            |             |             |          |
|-----------------|---------------|------------|-------------|-------------|----------|
| session_id (PK) | rider_id (FK) | started_at | device_type | app_version | ended_at |
|-----------------|---------------|------------|-------------|-------------|----------|

Rationale for Choosing Primary and Foreign Keys for the Table 2:

session\_id uniquely identifies one app visit and supports ordered journey analysis. rider\_id

is the foreign key to Rider because one rider can create many sessions. Device and app-version attributes belong at session grain instead of being redundantly repeated as rider attributes.

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**Table 3:**

Event

*(You may add more columns if necessary.)*

| event_id (PK) | session_id (FK) | event_time | event_type | page_type | neighborhood |
|---------------|-----------------|------------|------------|-----------|--------------|
|---------------|-----------------|------------|------------|-----------|--------------|

Rationale for Choosing Primary and Foreign Keys for the Table 3:

event\_id is the immutable unique event UUID and supports deduplication. session\_id links each event to its session, creating a one-to-many Session-to-Event relationship. Event time, type, page, and observed location belong at event grain; request\_car and begin\_ride event types support transactional funnel analysis.

## Section 3: Extraction and Transformation

Now that you have the requirements from your stakeholders, you want to understand the current state of what data is collected. That is how you recognize which additional data you need to achieve the future state. You ask the engineering team what data they are currently collecting in the pipelines and they provide you with section\_3\_event\_logs template (which you can download from the classroom) generated by rider's activities on the Flyber App. Also provided in the Project Resources.

### Extraction and Transformation-1

ETL is performed on the provided Event Logs Template and results will be transferred to the proposal template. The project's ETL should be created inside of your copy of the Event Logs template in the tab titled, ETL. Clicking on the link above will create a copy of the Event Logs for you

After being provided with a CSV log file, use extraction techniques to be able to get the data into a usable form. Because this needs to be a repeatable process we need to document it in order to assess its feasibility. Below,

1. Write the steps you took to extract the data and provide reasoning for why you used this method *Note: Don't forget to include any file type changes:*
2. Perform cleaning and transformation of the data in the ETL tab and document.
3. Document and provide rationale for all of your steps below as well.

Steps for Extraction:

*(You may add more steps if necessary.)*

1. **1. Preserve and inspect the source.** Used the supplied XLSX copy rather than editing the raw log in place; confirmed 124,980 event rows and the expected schema. XLSX preserves timestamp types and UUID text without CSV delimiter or encoding ambiguity.
2. **2. Standardize fields.** Converted Excel serial event\_time values to calendar dates, retained UUIDs as text, and standardized machine-readable category labels for presentation. This makes grouping reproducible while protecting identifier integrity.
3. **3. Apply the complete-day filter.** Limited reporting to October 5-11, 2019. A partial October 12 boundary segment was excluded because comparing an incomplete day with full days would create a false decline.
4. **4. Aggregate and validate.** Counted event\_uuid by date, event type, device type, page type, and neighborhood. Each categorical subtotal was reconciled to the daily total; all seven complete days passed.

## Transformation-2

Analyze the data from part 1 to answer the following questions:

1. How many events are being recorded per day?

| Date        | 10/5/2019 | 10/6/2019 | 10/7/2019 | 10/8/2019 | 10/9/2019 | 10/10/2019 | 10/11/2019 |
|-------------|-----------|-----------|-----------|-----------|-----------|------------|------------|
| Event Count | 9,891     | 18,056    | 18,202    | 17,963    | 17,600    | 17,694     | 17,595     |

2. How many events of each event type per day?

| Date        | 10/5/2019 | 10/6/2019 | 10/7/2019 | 10/8/2019 | 10/9/2019 | 10/10/2019 | 10/11/2019 |
|-------------|-----------|-----------|-----------|-----------|-----------|------------|------------|
| Choose Car  | 1,498     | 2,843     | 2,953     | 2,769     | 2,725     | 2,801      | 2,804      |
| Search      | 1,484     | 2,891     | 2,824     | 2,899     | 2,749     | 2,904      | 2,821      |
| Open        | 6,594     | 11,733    | 11,767    | 11,662    | 11,531    | 11,325     | 11,371     |
| Begin Ride  | 38        | 49        | 62        | 86        | 57        | 57         | 78         |
| Request Car | 277       | 540       | 596       | 547       | 538       | 607        | 521        |

3. How many events per device type per day?

| Date        | 10/5/2019 | 10/6/2019 | 10/7/2019 | 10/8/2019 | 10/9/2019 | 10/10/2019 | 10/11/2019 |
|-------------|-----------|-----------|-----------|-----------|-----------|------------|------------|
| ios         | 2,384     | 4,337     | 4,217     | 4,373     | 4,380     | 4,482      | 4,500      |
| android     | 1,463     | 2,870     | 2,854     | 2,729     | 2,744     | 2,562      | 2,672      |
| Desktop Web | 895       | 2,007     | 1,600     | 1,958     | 1,712     | 1,866      | 1,777      |
| Mobile Web  | 5,149     | 8,842     | 9,531     | 8,903     | 8,764     | 8,784      | 8,646      |

4. How many events per page type per day?

| Date        | 10/5/2019 | 10/6/2019 | 10/7/2019 | 10/8/2019 | 10/9/2019 | 10/10/2019 | 10/11/2019 |
|-------------|-----------|-----------|-----------|-----------|-----------|------------|------------|
| Search Page | 3,995     | 7,219     | 7,307     | 7,221     | 6,979     | 7,201      | 7,137      |
| Book Page   | 1,977     | 3,548     | 3,576     | 3,572     | 3,586     | 3,424      | 3,506      |
| Driver Page | 965       | 1,823     | 1,871     | 1,794     | 1,755     | 1,689      | 1,768      |
| Splash Page | 2,954     | 5,466     | 5,448     | 5,376     | 5,280     | 5,380      | 5,184      |

5. How many events for each location per day?

| Date      | 10/5/2019 | 10/6/2019 | 10/7/2019 | 10/8/2019 | 10/9/2019 | 10/10/2019 | 10/11/2019 |
|-----------|-----------|-----------|-----------|-----------|-----------|------------|------------|
| Manhattan | 6,869     | 12,591    | 12,807    | 12,180    | 12,270    | 12,371     | 12,201     |
| Brooklyn  | 2,009     | 3,737     | 3,590     | 4,025     | 3,440     | 3,400      | 3,556      |
| Bronx     | 250       | 533       | 507       | 469       | 510       | 394        | 558        |

|               |     |     |     |     |       |       |     |
|---------------|-----|-----|-----|-----|-------|-------|-----|
| Queens        | 595 | 842 | 905 | 893 | 1,026 | 1,069 | 936 |
| Staten Island | 168 | 353 | 393 | 396 | 354   | 460   | 344 |

### ETL Automation and Scalability:

Provide an analysis about this ETL process. Address and provide rationale for manually extracting, loading and transforming the data from the raw logs. Also address potential preliminary recommendations on improving this process.

The manual ETL was useful as an MVP validation because it exposed the schema, boundary-day issue, category cardinalities, and reconciliation rules. It was not efficient or scalable: repeated full-file extraction, spreadsheet grouping, and manual copying create latency, inconsistent logic, and error risk as event volume grows. Flyber should automate an incremental pipeline that lands immutable logs in cloud object storage, validates schema and required fields, deduplicates on event\_id, converts timestamps to UTC, quarantines invalid records, aggregates by partitioned event\_date, and loads warehouse tables on a schedule. Orchestration, monitoring, lineage, retries, and reconciliation tests should alert Engineering before downstream dashboards refresh.

## Section 4: Choosing Relevant Dataset

The previous exercise gave you a sneak peek into the Extraction and Loading aspects of ETLs in data pipelines. For making business decisions, a data consumer would like to have all the data they want. However, for any ecosystem, it is impossible to collect or provide everything that the customers need. In this exercise, you will get a taste of real world scenarios wherein:

- All the resources are not always available to get what you need.
- You have to get creative and get the most insights with a minimal data set.

Oftentimes your stakeholders/customers will “ask for the moon”, but you’ll have to push them to work with the small amount of information you have and get creative.

***Note: As you learned in the course, being a Data Project Manager involves an extraordinary amount of collaboration. Complete the next sections based on the following scenario.***

After the analysis in section 3, we made sense of the numbers, and realized the total number of events seems to be too small (this was a week’s worth of data, but you need at least a month). Further investigation reveals that this was a subset of logs, but the actual data that is being collected is much bigger. Working through this small data set was tedious, and repeating this exercise on a

much bigger data set manually won't be feasible. Considering the time constraints of this project, engineering is willing to help with some automation. They also have limited bandwidth and are busy scaling systems up.

Engineering is willing to provide some data, but they have asked for the criterion that is most important. To First provide your business question and provide a rationale for why this is the most important.

Choose one of the following prompts that you think can get you the most relevant information to proceed further.

1. How many events are being recorded per day?
2. How many events of each event type per day?
3. How many events per device type per day?
4. How many events per page type per day?
5. How many events for each location per day?

For your chosen question also answer the following using the data from section 3 to support your answer:

1. How much is the customer data increasing?
2. How much is the transactional data increasing?
3. How much is the event log data increasing?

Which of the following data is **most** important to answer this question? Why?

- Event Log Data
- Transactional Data
- Customer Data

Chosen criterion: How many events of each event type per day? This single dataset provides the most information because its row totals quantify event-log growth while the categories distinguish customer engagement (open, search, choose car) from transaction-intent and fulfilled-activity proxies (request car and begin ride). In the Section 3 sample, the category sum reconciles to total events every day, so the breakdown preserves the total and adds business meaning. Customer records themselves cannot be counted without a rider identifier or new-account event; therefore open/search/choose-car growth should be labeled a behavioral proxy, not a unique-customer count. Transactional growth is best approximated by request\_car and begin\_ride, while event-log growth is measured directly by the daily category sum. Event Log Data is consequently the most important source for this

question, supplemented later by Customer and Ride tables for exact unique-customer and financial transaction measures.

## **Section 5: [Optional] Loading and Visualization On Your Own**

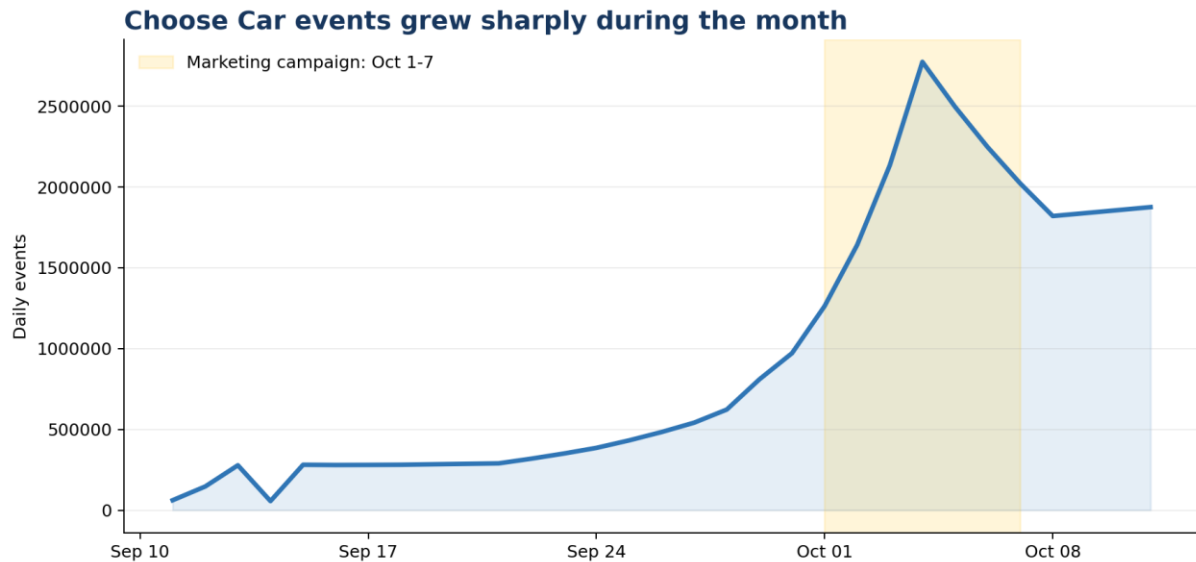
This section is an optional part of the project that you can do to make it stand out. We have provided visualizations in the appendix if you decide not to do this section. You can also use our visualizations to compare what you created

*After sharing your criterion with engineering, they give you a new set of data: Section 5 Event Type Log also available in the classroom resources. Also provided in the project resources section.*

Engineering provided you with the data you want, but you still have yet to achieve your ultimate goal as a Data Product Manager. Now, utilize the data to make business decisions. Your executives do not want you to give them a bunch of data tables; instead, they prefer visualizations to help convey the key insights succinctly. Visualizing this data will help you understand the underlying trends and help you determine the story that needs to be told in your proposal to executives.

In this section, you can load and visualize the data into whatever platform you would like. A Python Notebook, Tableau or any other visualization tool you are familiar with. Create two visualizations that might help you to better understand your data trends and place either a screenshot or exported image of your visualizations and the details of each below. Please provide the steps you took to visualize your data and what the visualization tells you about your data.

Visualization 1:



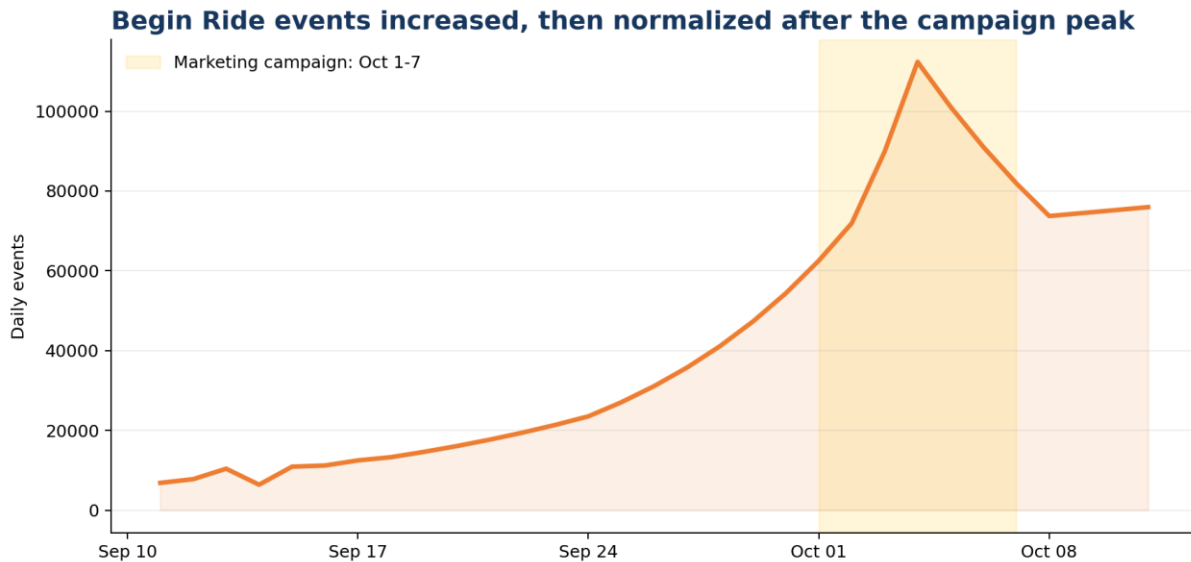
**Data Story:** This graph tells us:

Choose Car events rose from 61,416 to 1,875,536, a 30.54x increase. Growth accelerated into the October campaign, peaked on October 4, and stabilized at a level far above the beginning of the month. This is the fastest endpoint growth multiple among the five event types and indicates both stronger rider intent and substantial burst capacity requirements.

This graph was created using the following steps:

1. Loaded the Section 5 workbook and interpreted the first column as daily dates.
2. Selected Date and Choose Car, plotted a continuous line, formatted the y-axis as event count, and shaded October 1-7 as the campaign window.
3. Checked the first, final, and peak values against the source table and calculated final divided by initial volume.

Visualization 2:



**Data Story:** This graph tells us:

Begin Ride events increased from 6,838 to 75,956, an 11.11x increase. The series shares the broader campaign pattern but grows more slowly than upper-funnel events, suggesting that attention and ride consideration expanded faster than completed ride starts. Product and Operations should monitor conversion from Request Car to Begin Ride as scale increases.

This graph was created using the following steps:

1. Selected Date and Begin Ride from the same validated daily event-type dataset.
2. Created a continuous line with consistent date range, event-count units, and campaign-window shading.
3. Compared the first, final, and peak observations and interpreted the line as a lower-funnel operational measure.

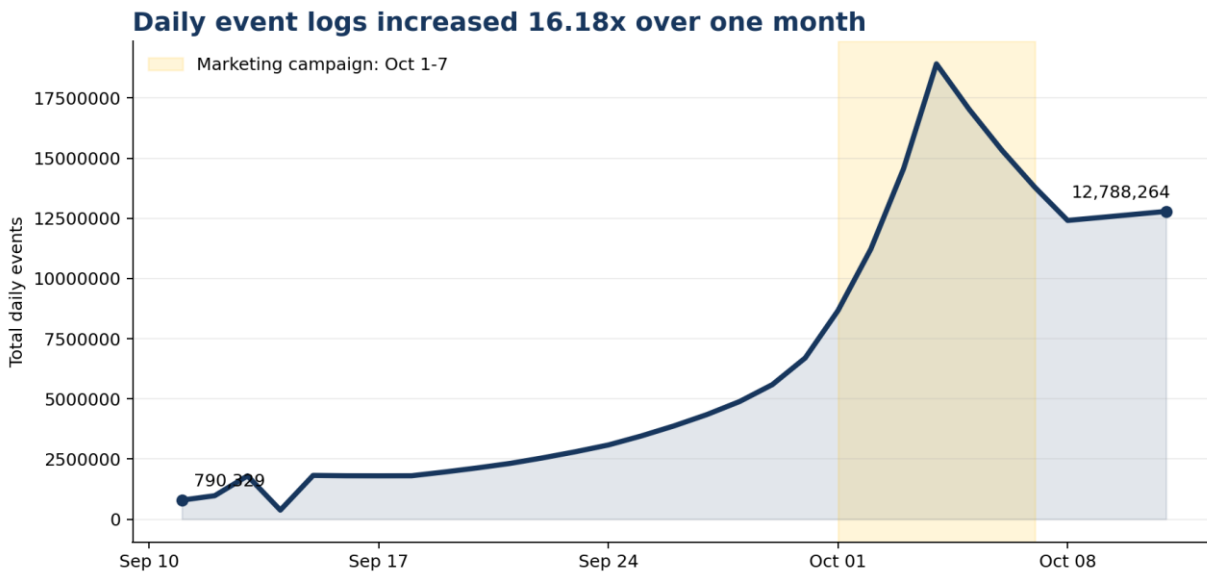
## Section 6: Business Insights

The Data is loaded and ready for analysis. We want to use this data as evidence to support our recommendations. It is important that we understand this data and the underlying trends and nuances that these visualizations show us. As you already know, any proposal backed up by data is always better received and considered more robust.

What is the story the data is telling you about Flyber's data growth? If you created Visualizations, you can use them as well, but they are not required). Include any data and calculations that were made to help tell that story and quantify the data growth.

## Data Growth for Last Month

Visualization:



Data and calculations used for quantifying of Flyber's Data Growth:

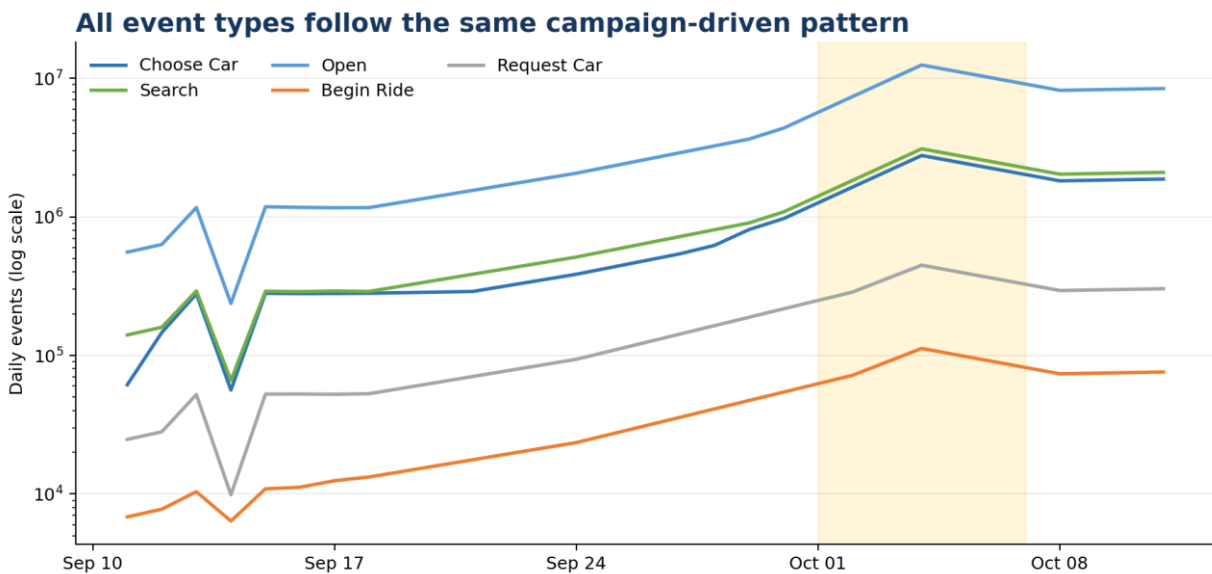
Daily total events increased from 790,329 on September 11 to 12,788,264 on October 11. Calculation:  $12,788,264 / 790,329 = 16.18x$ , equivalent to an absolute increase of 11,997,935 events or 1,518.1%. Volume peaked at 18,918,096 on October 4. Because the calculation compares daily run rates exactly one month apart, it measures growth in operational load without conflating it with the number of days in a period.

What is the fastest growing data and why?

Event logs are the fastest-growing data class in absolute volume because every rider journey produces multiple events: open, search, choose car, request car, and begin ride. Among observed event types, Choose Car has the fastest endpoint growth at 30.54x; Open remains the largest volume at 8,438,774 events on October 11. Customer data should grow closer to unique rider acquisition, while transactional data grows with requests and rides and therefore produces fewer rows than behavioral telemetry. Exact cross-class rates require unique rider and ride records; the event dataset supports this ordering through event-type proxies, not exact customer-table counts.

## All Event Type Data

Visualization:



What is the Data Story our data tells for each of the following:

- Graph Pattern
- Good or Bad
- October Marketing Campaign
- Marketing Campaign Impact
- Importance of Relationship Between Marketing Campaigns and Data Generation

**Graph pattern:** All five series broadly rise together, peak on October 4, decline through October 8, and then stabilize or increase slightly. The logarithmic scale shows that the shared shape persists despite order-of-magnitude differences in volume. **Good or bad:** synchronized growth is positive evidence of end-to-end engagement, but upper-funnel Choose Car growth outpaces Begin Ride growth, so conversion and supply constraints require monitoring. **October campaign:** the October 1-7 campaign coincides with rapid acceleration and the peak; average total volume during the campaign was 14.22 million events per day versus 4.56 million during September 24-30, a 211.6% increase. **Impact:** campaign activity materially increases both business demand and telemetry load, although the observational data alone does not prove causality. **Importance:** Marketing, Engineering, Product, and Operations must jointly forecast campaign traffic, pre-scale ingestion and app infrastructure, monitor conversion, and distinguish durable post-campaign growth from a temporary spike.

## Section 7: Data Infrastructure Strategy

Thus far we have:

- identified data stakeholders and their data needs.
- Identified what data is currently being collected and what data needs to be collected.
- Identified data insights and growth trends.

Now, it's time to tie all the loose threads together and bring this process to its logical conclusion by suggesting which Data Warehouse (DWH) Flyber should invest in and why. Using data warehouse options below, suggest whether Flyber should choose an on-premise or Cloud data warehouse system and which specific data warehouse would best serve Flyber's data needs.

### **Data Warehouse Options:**

Cloud:

- Amazon Redshift
- Google BigQuery
- Snowflake
- Microsoft Azure

On-Premise:

- Oracle Exadata
- Teradata, Vertica
- Apache
- Hadoop

You will address the following factors with a rationale as to why the DWH chosen is the best for Flyber:

- Cost
- Scalability
- In-house Expertise
- Latency/Connectivity
- Reliability

### **Cloud vs On-Premise**

Provide an evidence based solution as to why Flyber would be best served by a Cloud or on-premise DWH. In this response, you don't need to specify *which* specific Cloud or on-premise DWH product you will choose, just if it will be Cloud or on-premise. Remember to address the factors above.

Recommendation: use a cloud data warehouse. Cost: a startup avoids large capital purchases and can align spending with storage and query use, with budgets, retention tiers, and workload controls. Scalability: the observed 16.18x daily event growth and campaign peak require elastic compute and date-partitioned storage without lengthy hardware procurement. Expertise: managed services reduce database administration and allow the small team to focus on data models, quality, and products, although SQL, governance, security, and FinOps skills remain necessary. Latency/connectivity: a cloud warehouse supports frequent batch or near-real-time analytics close to cloud object storage; the live application should continue to use an operational database rather than querying the warehouse. Reliability: multi-zone managed infrastructure, backups, monitoring, and service-level commitments are stronger and faster to operationalize than a startup-owned

cluster. An on-premise system would create fixed capacity, procurement delay, and specialized operating burden precisely when traffic is volatile.

### **Suggested DWH**

Provide an evidence based solution as to which DWH product is best for Flyber. Remember to address the factors above.

Suggested DWH: Google BigQuery. BigQuery is a serverless cloud warehouse suited to rapidly growing event data because compute scales without cluster sizing, storage and compute are separated, and standard SQL supports Engineering, Product, Marketing, Finance, and BI users. Flyber should partition fact\_event by event\_date, cluster by event\_type and session\_id or rider\_id, materialize frequently used daily aggregates, and apply workload labels, query quotas, budgets, and lifecycle rules to control cost. Streaming or micro-batch ingestion can support low-latency operational dashboards, while scheduled transformations provide stable executive metrics. Built-in replication, managed maintenance, encryption, access controls, audit logs, and integration with cloud storage reduce the reliability and expertise burden. This recommendation follows directly from the correlated campaign spike, 18.9 million-event peak day, and 16.18x monthly run-rate growth; it provides elasticity while preserving a normalized core model and governed analytical marts.

## Image Appendix

Image 1: Log Growth

Log Growth

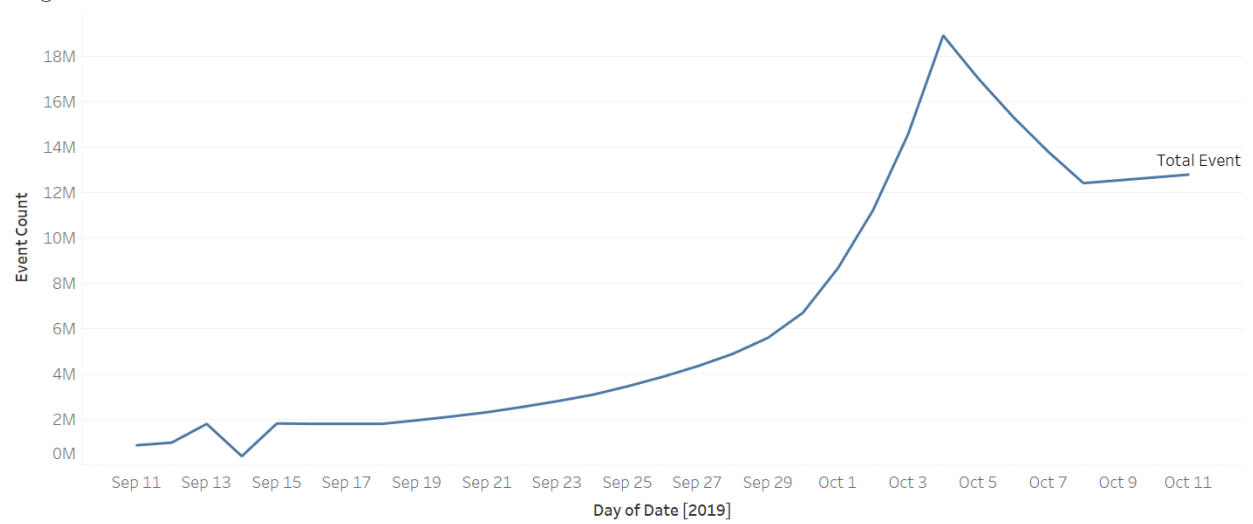


Image 2: Ride Growth

Ride Growth

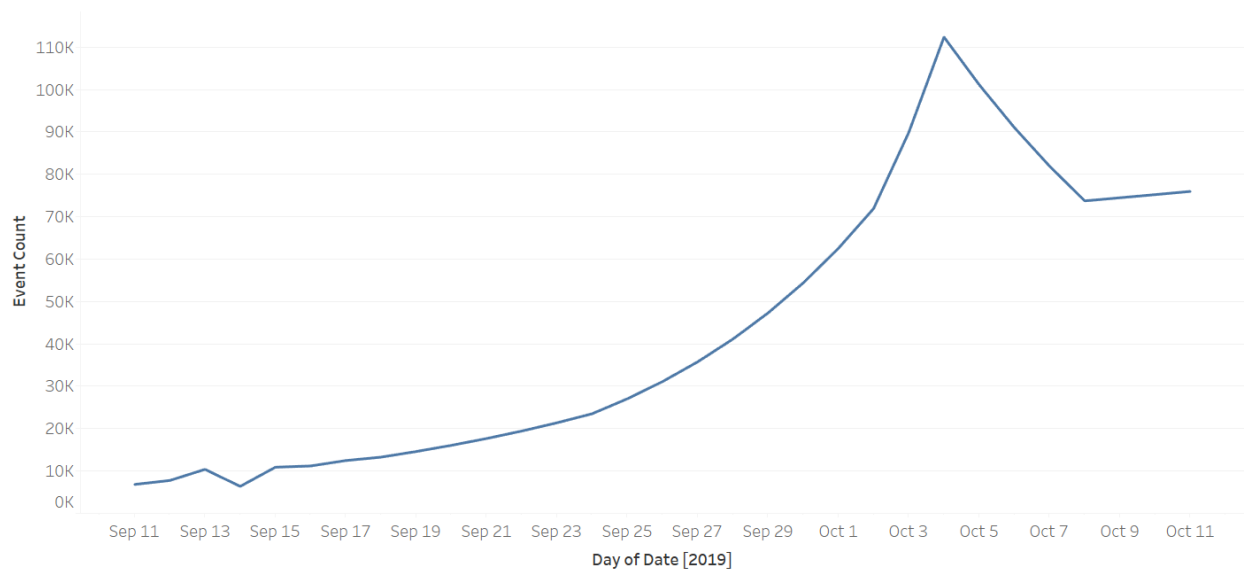


Image 3: Total Event Count

Total Event Count

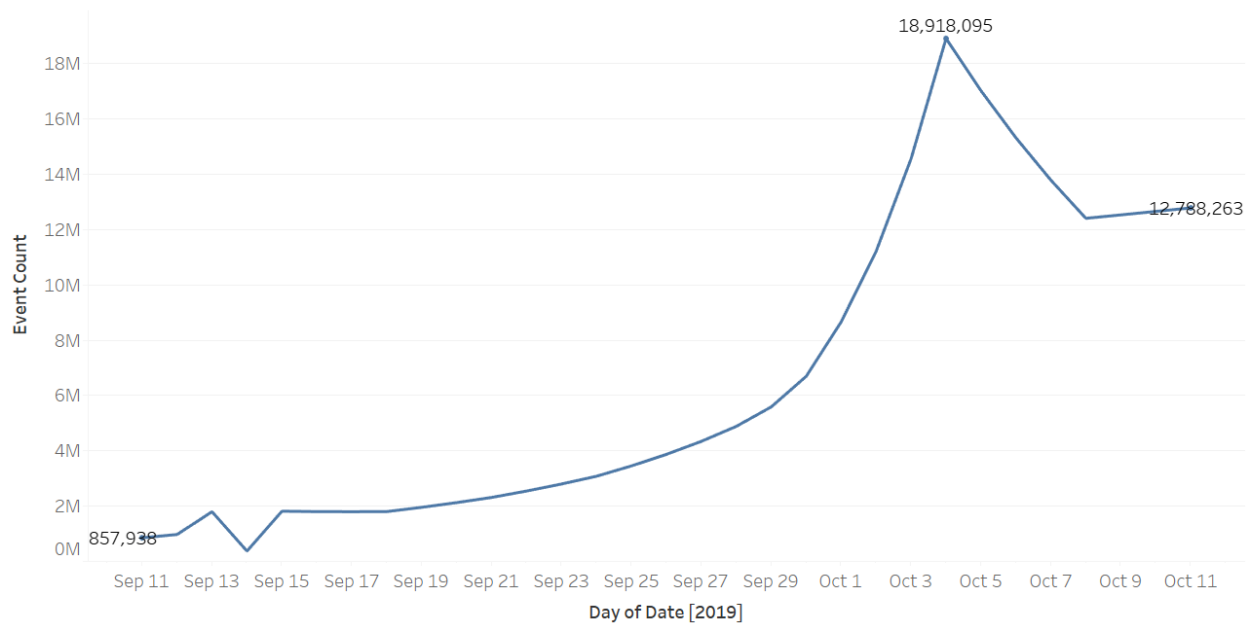


Image 4: All Events Log Scale

All Types of Events on a Logarithmic Scale.

